Introduction to Software Engineering

Project Idea & Team Contract



Software Engineering Department
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Project Idea & Team Contract

Objectives

This document focus on the following topics:

- ✓ Describing the project idea at a high level so the teaching staff can approve it
- ✓ Agreeing on how the team will work together (team contract)
- ✓ PA0 is not graded, but it is required before the team starts PA1

1 Team Information

Item	Value
Team ID	T01
Team name	The 4 idiots
Team lead	Nguyễn Lương Hoàng Duy
Communication channel	Zalo, google meet

Student ID	Full name	Email	Main role	Backup role
24127352	Nguyễn Lương Hoàng Duy	nlhduy2433@clc.fitus.edu.vn	Team lead	Developer
24127343	Trần Phúc Đạt	tpdat2410@clc.fitus.edu.vn	Designer	Developer
24127436	Niê Ngọc Lâm	nnlam2427@clc.fitus.edu.vn	Developer	Business analyst
24127521	Nguyễn Việt Quang	nvquang2410@clc.fitus.edu.vn	Tester	Designer

[Roles may change later; confirm the final assignment in PA1. Every role needs one main person and one backup.]

2 Project Idea Overview

[About 1–2 pages. Keep it high level; detailed requirements come in PA2.]

2.1 Project Name and One-Sentence Pitch

"WhatYaDoing? – a desktop application that helps users plan their days, prioritize tasks and track their daily progress."

2.2 Problem and Motivation

- **Problems the system solves**:
  - **Lack of Daily Structure:** Individuals often start their day without a clear, organized list of tasks, leading to loss of focus, poor time management, and reduced productivity.
  - **Cognitive Overload:** Relying on mental memory to keep track of multiple tasks, deadlines, and progress causes mental fatigue, anxiety, and a higher risk of forgetting important obligations.
  - **Task Prioritization Struggles:** Without visual organization, users could struggle to differentiate between urgent responsibilities and low-priority tasks, making it difficult to decide what to tackle next.
- **Who has the same problems**:
  - **Students:**
    - *Needs:* Managing assignment deadlines, study schedules, course projects, and daily personal chores across various subjects.
    - *Pain Point:* Balancing multiple overlapping assignments without clear progress visibility.
  - **Remote Workers & Freelancers:**
    - *Needs:* Self-regulating working hours, organizing daily deliverables, and separating daily routines from long-term projects.
    - *Pain Point:* Working without direct supervision requires strict personal task management to stay on track.
  - **Office Professionals & Knowledge Workers:**

- *Needs:* A quick, offline-accessible desktop tool to dump immediate daily tasks without relying on company provided tools.
- *Pain Point:* Spending too much time navigating complex software just to track simple daily to-do items.

2.3 Target Users

User group	What they need from the system
Students	A free, locally stored application to manage day-to-day tasks and track academic progress.
Office workers	An intuitive, user-friendly interface to quickly plan and structure their workday without steep learning curves.
Freelancers	Productivity features such as priority-based organization, task timers, and daily completion summaries.

2.4 Main Features

[List 5–10 main features. Mark the ones you consider Must-have; only Must-have features must be fully implemented by PA5.]

#	Feature	Short description	Must-have?
1	Task Creation and Editing	Allows users to create, edit, delete, and mark daily to-do tasks as completed.	Yes
2	Local Data Storage	Automatically saves all data on tasks and app settings locally on the user's computer to ensure offline access and privacy.	Yes
3	Priority and Coloring Coding	Enables users to categorize tasks by priority level (e.g., High, Medium, Low) with visual color tags.	Yes
4	Daily Task List and Filtering	Displays daily tasks cleanly and allows filtering by status (All, Active, Completed) or priority.	Yes
5	Task Progress	Provides a visual bar or percentage showing daily task completion progress	Yes

6	Task Timer	Includes a customizable timer to help users focus on specific tasks for set durations.	No
7	Daily summary	Generates a daily summary report compiling all completed tasks at the end of the day.	No
8	Search and Quick Keyword Filter	Allows users to search through their task list using keywords.	No

2.5 Similar Products

Microsoft To Do

Microsoft To Do is a task management application that allows users to create daily task lists, set due dates, organize tasks, and mark tasks as completed. Similar to WhatYaDoing?, it helps users organize their daily activities and keep track of their progress. However, WhatYaDoing? focuses on providing a lightweight desktop application with locally stored data, making it suitable for users who prefer a simple tool that works offline without relying on an online account.

Todoist

Todoist is a task management application that supports task organization, priorities, due dates, and productivity tracking. It is similar to WhatYaDoing? because both applications help users prioritize tasks and manage their workload. However, WhatYaDoing? aims to provide a simpler interface with essential daily planning features, including color-coded priorities, task filtering, and a visual daily progress indicator, without the complexity of a more comprehensive task management platform.

2.6 Platform and Technology Ideas

- **Platform**: Desktop.
- **Main Programming language**: Python.
- **Data storage**: Json files.

2.7 Feasibility, Scope and Open Questions

[Can a team of 4–5 students build the Must-have features in about three weeks of coding (weeks 8–10)? List what is out of scope, the main risks and the questions you want the TA to answer.]

According to our initial evaluation, implementing the listed must-have features are possible in the span of 3 weeks. This however, does not mean that there are no risk, we can still run into unexpected problems such as:

- A team member is suddenly unavailable for substantial amount time during the 3 weeks of implementing. For reasons such as sickness, family emergencies or being overwhelmed by other courses that they are taking, a member can suddenly not be able to contribute to the project, this is especially impactful as the team size is rather small.
- Bad initial evaluation, we determine that implementing core features are possible in 3 weeks based on our initial evaluation. This however, could be wrong, leading to unexpected difficulties down the line.

3 Team Contract

[Discuss each item as a team and write down what you agree. The contract is reviewed when a conflict occurs and is used as evidence when the teaching staff adjust individual contribution factors.]

3.1 Goals and Expectations

The team's target grade is in the range of: 7 – 8, therefore, every team member is expected to dedicate at least one hour of their day or at least 7 hours per week to the project.

3.2 Communication

- **Main channel for discussions:** The team's Zalo group.
- **Main channel for team meetings:** Google meet, the link for the meeting will be sent to the team's Zalo group 10 minutes before the meeting begins.
- **Expected response time to messages:** Within 24 hours.
- **Where decisions and files are kept:** Shared google drive and git repository.

3.3 Meetings

- **Weekly status meeting with the TA:** 2 P.M on Saturday using google meet.
- **Internal team meetings:** Every saturday at 8 P.M using google meet.
- **Attendance rule:** Inform the team lead in advance if you cannot attend the meeting, missed meetings are recorded in the weekly status report.
- **Weekly status report deadline chosen by the team:** Each member is expected to have given a status report by Saturday every week.

3.4 Roles and Responsibilities

Aside from the roles in the team table above, each member must fulfill additional duties as follows:

Name	Additional duties
Nguyễn Lương Hoàng Duy	Prepare as the main speaker for any presentations, meetings with the TA./Teacher.
Trần Phúc Đạt	Prepare the slides and other documents for any presentations, meetings.

Niê Ngọc Lâm	Prepare the slides and other documents for any presentations, meetings.
Nguyễn Việt Quang	Help the team lead with reviewing the documents (PA, Report, etc). Keep the AI usage log for each PA.

3.5 Work Standards

- **Internal deadline:** Each member must finish their work at least one day before the PA deadline.
- **Quality:** Every document and pull request are reviewed by at least one other member.
- **Tool:** Every task is on the task board with an owner and due date; code goes through pull requests

3.6 Use of AI Tools

- **Permitted Tools:** Team members are allowed to use AI assistants (such as Gemini, ChatGPT, and Claude) for brainstorming, drafting text, writing boilerplate code, debugging, and reviewing sections of the project report.
- **Review and Responsibility:** All AI-generated outputs, code snippets, and text must be thoroughly reviewed, tested, and understood by the assigned team member before being integrated into the project. No raw, unverified AI output should be submitted directly.
- **Academic Integrity & Transparency:** Team members must not use AI tools to bypass core learning tasks or write entire assignments without human oversight. Any significant use of AI must be fully disclosed and documented in the AI Usage Log at the end of the document.
- **AI Log Management:** Nguyễn Việt Quang is responsible for keeping and updating the AI usage log for each PA based on inputs from the team.

3.7 Decision Making

- **Consensus First:** The team will first attempt to reach consensus through open discussion.
- **Majority Voting:** If consensus cannot be reached, decisions will be made by a majority vote. Each member has one vote per proposal. A majority vote requires at least 3 active voters to proceed; otherwise, discussion continues.

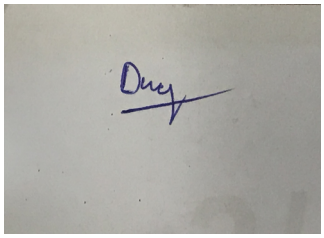
- **Abstentions:** Members may choose to abstain for reasons such as lacking knowledge or confidence in a proposal.
- **Tie-Breaker Process:** In the event of a tie, the team will hold one brief follow-up discussion focusing on compromise or alternative solutions, followed by a final vote. If a tie remains unresolved, the **Team Lead** will make the final decision to keep the project moving forward.

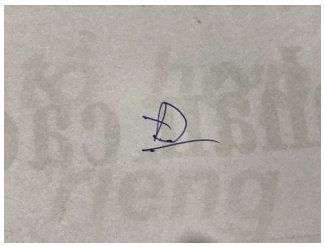
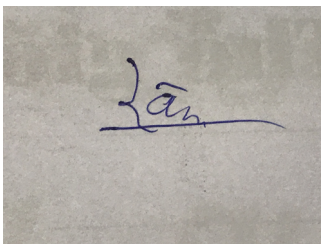
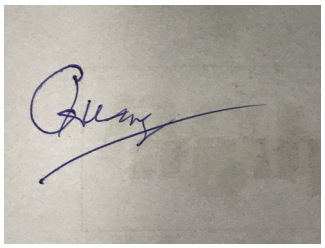
3.8 Handling Conflicts and Missed Work

- **Step 1: Direct 1-on-1 Discussion:** The affected team member(s) and Team Lead (or peer) will discuss the issue privately to identify root causes, adjust the plan if needed and set a new deadline.
- **Step 2: Team Workload Realignment:** If the issue persists or impacts broader project milestones, the issue will be brought to a team meeting. Discussion will be held on how to proceed, possible fixes and re-assigining tasks to ensure project goals are met.
- **Step 3: Formal Escalation (TA):** If a team member repeatedly fails to meet agreed deadlines or uncooperative conflict hinders team performance despite Steps 1 and 2, the team will inform the TA. The issue will be documented in the weekly status report and factored into individual contribution evaluation.

3.9 Signatures

[By signing, each member agrees to follow this contract.]

Student ID	Full name	Signature	Date
24127352	Nguyễn Lương Hoàng Duy		13/10/2026

24127343	Trần Phúc Đạt		13/10/2026
24127436	Niê Ngọc Lâm		13/10/2026
24127521	Nguyễn Việt Quang		13/10/2026

4 AI Usage Log

[Required in every PA (course GenAI policy). List each significant use of AI tools. Write "No AI tools were used" if none. Undisclosed AI use is treated as an academic integrity violation.]

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Lương Hoàng Duy	Gemini	Evaluating section 3.7 and 3.8, to see if the handlings of decision making and conflict resolution are sound or not. Provide fixes or alternatives.	Several issues with the darft were found and fixes proposed by Gemini were implemented.
2	Niê Ngọc Lâm	ChatGPT	Drafting and refining section 2.5	Read and verified the AI comparisons
3	Nguyễn Việt Quang	Claude	Evaluating the complexity of the project and expected outcome for a fair and logical workload distribution among team members.	Read the previous distributed-work and changes it to satisfy each members ability.
4	Trần Phúc Đạt	Gemini	Assisting with drafting and refining Section 3.6 (Use of AI Tools), as well as formatting and structuring the project plan overview and UI	Reviewed the AI usage guidelines and suggested UI layouts, customizing them to align with our team policies and desktop application requirements.

			layout design ideas. .	